

ABERPORTH, United Kingdom

WMO: 035020

Lat: 52.139N

Lon: 4.571W

Elev: 134

StdP: 99.73

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.4	-0.1	-5.9	2.3	0.3	-4.4	2.6	1.3	20.6	6.9	19.0	7.3	5.6	150	0.763

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	22.1	16.6	20.2	16.0	18.8	15.5	17.7	20.3	16.9	19.0	16.2	18.0	5.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.2	18.5	16.1	11.6	17.7	15.4	11.1	17.1	50.4	20.4	48.0	19.0	46.0	18.2	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.1	13.5	-3.1	26.3	1.4	2.4	-4.1	28.0	-4.9	29.4	-5.7	30.8	-6.8	32.6		
(5)			-4.3	19.9	1.4	1.3	-5.3	20.8	-6.1	21.5	-6.9	22.2	-8.0	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	5.6	5.6	6.7	8.4	10.9	13.4	15.1	15.3	13.8	11.4	8.4	6.4		
	DBStd	4.25	2.71	2.56	2.46	2.44	2.46	2.24	2.00	1.84	1.85	2.22	2.39	2.90		
	HDD10.0	624	136	124	105	59	16	1	0	0	0	11	58	114		
	HDD18.3	3009	393	356	361	299	230	149	103	98	137	215	298	370		
	CDD10.0	666	1	1	3	11	45	103	158	164	113	54	10	2		
	CDD18.3	9	0	0	0	0	1	2	3	3	1	0	0	0		
	CDH23.3	32	0	0	0	0	3	7	10	12	1	0	0	0		
	CDH26.7	4	0	0	0	0	0	1	1	2	0	0	0	0		
(14) Wind	WSAvg	6.7	8.4	8.1	7.1	6.2	5.7	5.4	5.2	5.6	6.2	7.2	7.7	8.1		
(15) Precipitation	PrecAvg	1251	144	101	77	76	76	74	80	92	86	143	155	147		
	PrecMax	1599	253	258	150	159	181	180	195	211	142	274	283	300		
	PrecMin	982	6	27	25	18	11	21	18	6	14	47	79	25		
	PrecStd	152	58	53	35	35	36	42	41	47	31	57	55	68		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	12.2	15.4	19.0	21.7	24.2	24.9	24.6	21.7	18.2	14.4	12.8		
		MCWB	10.7	10.1	10.5	12.8	15.2	17.9	17.9	18.0	16.8	14.6	13.0	12.0		
	2%	DB	11.0	10.8	12.2	15.1	18.4	20.6	21.6	21.2	19.3	16.3	13.4	12.0		
		MCWB	10.3	9.6	9.5	10.7	13.8	15.7	16.6	17.0	15.5	14.4	12.5	11.4		
	5%	DB	10.2	10.0	11.0	12.9	16.2	18.4	19.8	19.3	17.8	15.3	12.6	11.2		
		MCWB	9.3	8.9	9.3	10.0	12.7	14.7	16.2	16.2	15.2	13.8	11.7	10.4		
	10%	DB	9.4	9.2	10.1	11.7	14.5	16.8	18.4	18.1	16.7	14.6	11.9	10.4		
		MCWB	8.6	8.2	8.8	9.4	11.9	14.0	15.6	15.7	14.6	13.2	10.8	9.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.0	11.2	13.0	16.2	18.5	18.6	19.1	17.9	15.9	13.7	12.2		
		MCDB	11.6	11.5	13.0	17.9	20.5	22.9	22.6	22.8	20.3	17.0	14.0	12.4		
	2%	WB	10.4	10.0	10.4	11.5	14.5	16.3	17.6	17.8	16.6	15.0	12.8	11.5		
		MCDB	10.9	10.5	11.5	13.9	17.4	19.2	20.3	19.8	18.2	15.9	13.2	11.9		
	5%	WB	9.6	9.2	9.7	10.6	13.3	15.3	16.7	17.0	15.8	14.2	11.9	10.7		
		MCDB	10.1	9.8	10.6	12.4	15.4	17.5	18.9	18.5	17.0	15.1	12.5	11.2		
	10%	WB	8.8	8.4	8.9	9.8	12.4	14.5	16.0	16.2	15.1	13.5	11.1	9.8		
		MCDB	9.4	9.2	10.0	11.4	14.1	16.4	17.8	17.7	16.4	14.4	11.7	10.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.0	4.6	5.1	5.4	5.3	5.1	5.0	4.9	4.4	3.8	3.8		
		MCDBR	4.3	4.4	5.6	6.9	7.9	7.9	7.3	7.0	6.7	4.9	4.0	4.1		
	5% WB	MCWBR	4.6	4.0	4.0	4.4	4.8	4.8	4.1	4.0	4.2	3.9	4.1	4.6		
		MCWBR	4.4	4.2	5.1	6.2	6.9	6.9	6.4	6.0	5.4	4.5	3.9	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.337	0.363	0.378	0.385	0.375	0.379	0.363	0.357	0.346	0.324	0.314		
		taud	2.335	2.388	2.336	2.320	2.343	2.395	2.388	2.449	2.460	2.448	2.391	2.310		
	Ebn at Noon		664	777	825	857	867	878	868	865	827	757	664	603		
		Edh at Noon	63	79	100	114	117	112	111	100	88	73	59	55		
(44) All-Sky Solar Radiation	RadAvg		0.72	1.47	2.71	4.29	5.33	5.79	5.24	4.39	3.21	1.78	0.89	0.54		
	RadStd		0.07	0.13	0.21	0.30	0.35	0.49	0.46	0.29	0.22	0.15	0.08	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page